

9.6.20

Suppose the camera shop in exercise 4 can obtain at most ten A76 batteries but can get at least 30 of each of the other types.

- a. How many ways can a total inventory of 30 batteries be distributed among the eight different types?
- b. Suppose that in addition to being able to obtain only ten A76 batteries, the store can get only six of type D303. How many ways can a total inventory of 30 batteries be distributed among the eight different types?

- a. How many ways can a total inventory of 30 batteries be distributed among the eight different types?

The problem states that A76 can appear at most 10 times. So these are allowed:

$$0, 1, 2, \dots, 9, 10$$

These are forbidden:

$$11, 12, 13, \dots, 30$$

We want to count the forbidden group so that we can subtract it from all possible inventories. To represent every forbidden inventory, we say that **every forbidden inventory has at least eleven A76 batteries**. Hence:

$$30 - 11 = \boxed{19}$$

Those remaining 19 can still be any of the eight types, including A76. Hence:

$$\binom{8 + 19 - 1}{19} = \binom{26}{19} = \frac{26!}{19!(26 - 19)!} = \frac{26!}{19!7!} = \boxed{657,800}$$

Let A be the set of every forbidden inventory that has at least eleven A76 batteries. Then:

$$N(A) = 657,800$$

Let S be the set of every inventory that has 30 batteries of eight types. Then:

$$N(S) = \binom{37}{30} = \frac{37!}{30!(37 - 30)!} = \frac{37!}{30!7!} = \boxed{10,295,472}$$

$$N(S) - N(A) = 10,295,472 - 657,800 = \boxed{9,637,672}$$

Solution: a. There're 9,637,672 ways a total inventory of 30 batteries can be distributed among the eight different types, if the camera shop can obtain at most ten A76 batteries.

b. Suppose that in addition to being able to obtain only ten A76 batteries, the store can get only six of type D303. How many ways can a total inventory of 30 batteries be distributed among the eight different types?

By the logic above, we can do the same calculation for the constraints introduced for D303, where the forbidden case contains at least seven D303 batteries:

$$\binom{8+23-1}{23} = \binom{30}{23} = \frac{30!}{23!(30-23)!} = \frac{30!}{23!7!} = \boxed{2,035,800}$$

Let B be the set of every forbidden inventory that has at least seven D303 batteries. Then:

$$N(B) = 2,035,800$$

$$N(S) - N(A) - N(B) = 10,295,472 - 657,800 - 2,035,800 = \boxed{7,601,872}$$

But some inventories are in both A and B , meaning they have at least eleven A76 and at least seven D303, which means we subtracted those twice. Hence, we must add $N(A \cap B)$ back to the total ways.

$$N(A \cap B) = \binom{8+12-1}{12} = \binom{19}{12} = \frac{19!}{12!(19-12)!} = \frac{19!}{12!7!} = \boxed{50,388}$$

$$7,601,872 + 50,388 = \boxed{7,652,260}$$

Solution: b. There're 7,652,260 ways a total inventory of 30 batteries can be distributed among the eight different types, if the camera shop can obtain at most ten A76 batteries and at most six D303 batteries.